

Adverse childhood experiences are associated with prediabetes diagnosis in college students

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Abstract

An estimated 24% of young adults have prediabetes, making it an important concern in the college student population. Guided by prior findings on the association between adverse childhood experiences (ACEs) and Type 2 diabetes, we examine the relationship between ACEs and self-reported prediabetes. We demonstrate that individuals with four or more ACEs are over 200% more likely to have a prediabetes diagnosis, a value on par with established CDC self-report measures of prediabetes risk.

Introduction

An estimated 37.3 million people in the US have diabetes, a condition ranked as the 7th leading cause of death in this country. (National Diabetes Statistics Report 2022) It is therefore critical to identify factors that contribute to the development of this chronic health condition. One area that has recently received attention is the association between adverse childhood experiences (ACEs) and diabetes. It is proposed that understanding the role of ACEs in diabetes development might lead to improved diabetes management (Ittoop et al., 2020) and potentially facilitate the development of early preventative interventions for children with ACEs (Campbell et al., 2018). Although there is some evidence for the ACEs-diabetes association (Huffhines et al., 2016), research on prediabetes is limited. Prediabetes refers to a condition in which blood sugar levels are higher than normal but not at the threshold for diabetes. Prediabetes is a risk factor for the development of diabetes, with 5-10% of pre-diabetics becoming diabetic every year (Tabák et al., 2012). An estimated 24% of young adults have prediabetes (Andes et al., 2020), making it an important concern in the college student population, particularly given students' tendency to underestimate their risk of developing diabetes (Antwi et al., 2020). However, research on ACEs and prediabetes in college-aged populations is limited. (Brody et al., 2017; Campbell et al., 2018). To further investigate this issue, we examined the association between ACEs and prediabetes diagnosis in a predominantly Hispanic sample of undergraduates.

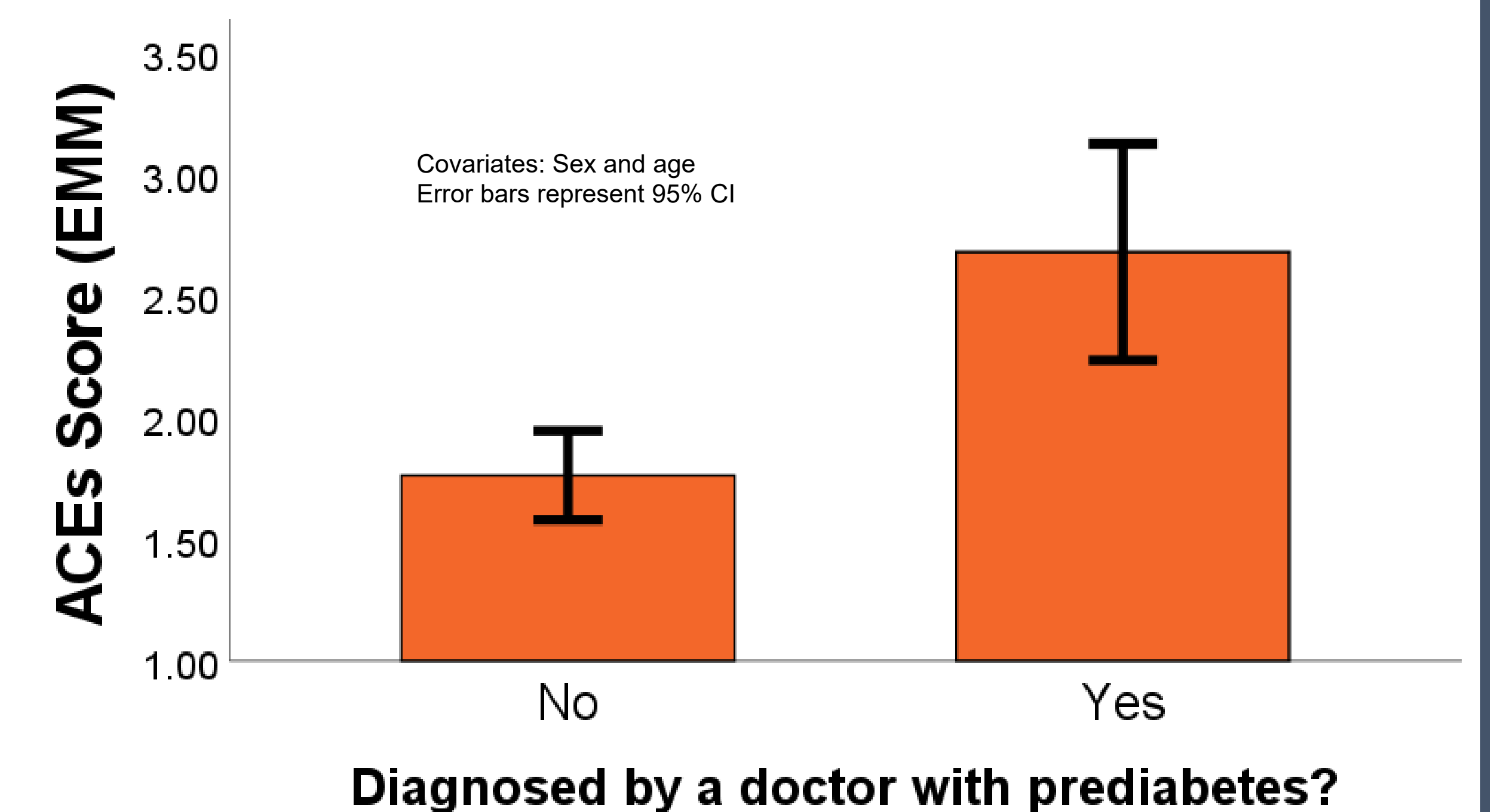
Methods

Five hundred and twenty-nine undergraduates (77.3% female; 94.5% Hispanic; mean age 19.65) completed an online anonymous survey that included a measure of ACEs (Adverse Childhood Experiences Questionnaire) and the Centers of Disease Control and Prevention (CDC) Prediabetes Risk Test, along with a separate self-report question assessing whether the participant had been diagnosed with prediabetes by a doctor (based on blood test results).

Results

Approximately 14% of the sample indicated that their doctor had previously diagnosed them with prediabetes based on a blood test result, and 9.2% of the sample met the threshold for increased prediabetes risk, based on their CDC Prediabetes Risk Score. Only six participants (1.1%) indicated that they had previously been diagnosed with Type 2 diabetes. Sixty-eight percent of the sample reported having experienced one or more ACEs and 21% reported having experienced four or more.

A logistic regression was conducted to examine whether those with four or more ACEs (a commonly used threshold) had a higher risk of self-reported prediabetes. Controlling for age and sex, individuals with four or more ACEs were 202% more likely to report a diagnosis of prediabetes than those with fewer than four (OR = 3.02, 95% CI = 1.75 – 5.20, $p < .001$). Those with a Prediabetes Risk Score of five or more (the recommended threshold for increased prediabetes risk) were 227% more likely to report the diagnosis of prediabetes than those with a score of fewer than five (OR = 3.27, 95% CI = 1.58 – 6.73, $p < .001$). We also examined whether ACEs scores were higher in those previously diagnosed with prediabetes. An ANCOVA, controlling for age and sex, revealed significantly higher ACEs in those reporting prediabetes ($M = 2.68$) than those who did not ($M = 1.76$), $F(1,508) = 13.98$, $p < .001$.



Discussion

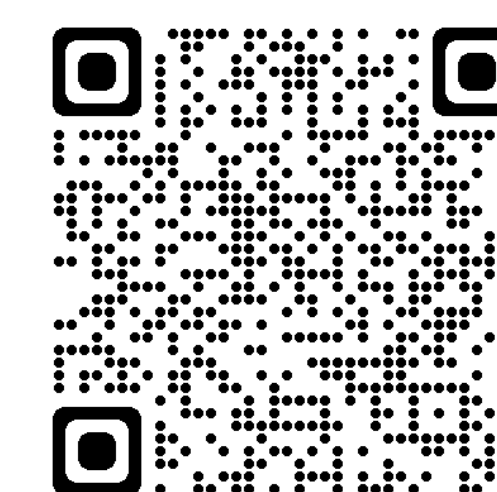
The prevalence of self-reported prediabetes within the investigated sample (13%) closely matches previously reported estimates of prediabetes awareness in US adults, especially in minority race-ethnicity groups. Our findings build upon existing studies demonstrating an association between ACEs and T2DM diagnosis, providing additional evidence that this association can be observed at earlier stages, therefore, improving the diagnosis process and developing more effective treatment plans for their patients.

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